

Application Deployment

Cloned Repository

```
root@ip-172-31-35-199:~# git clone https://github.com/Vennilavan12/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 8, done.
remote: Total 8 (delta 0), reused 0 (delta 0), pack-reused 8 (from 1)
Receiving objects: 100% (8/8), 100.04 KiB | 14.29 MiB/s, done.
root@ip-172-31-35-199:~# ls
Brain-Tasks-App  snap
root@ip-172-31-35-199:~# cd Brain-Tasks-App/
```

Build Docker Image

```
root@ip-172-31-35-199:~/Brain-Tasks-App# docker build -t brainstack:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  489.5kB
Step 1/6 : FROM nginx:alpine
alpine: Pulling from library/nginx
589002ba0eae: Pulling fs layer
```

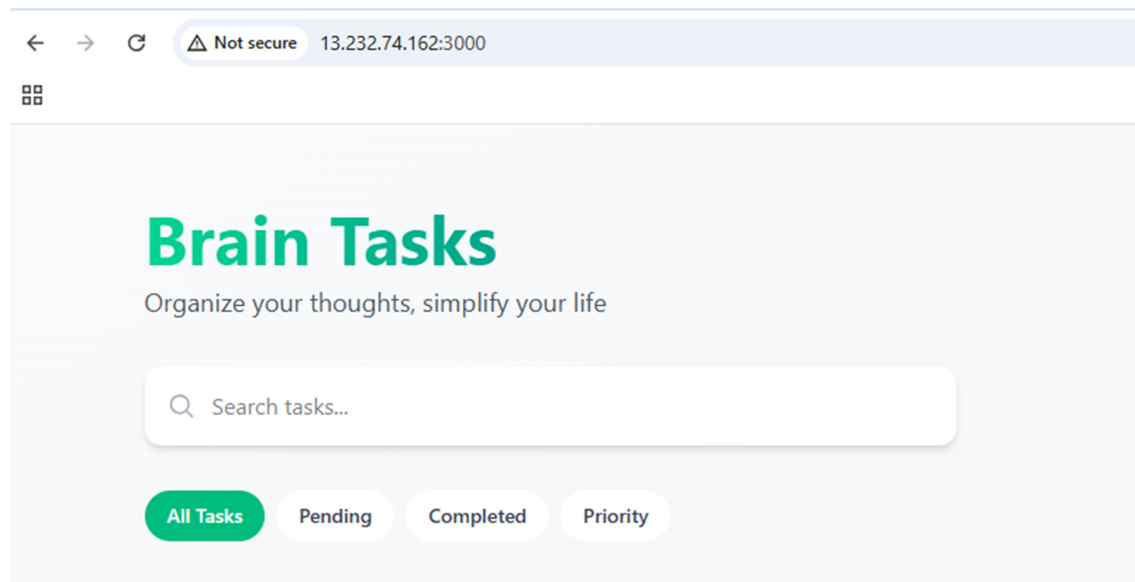
Docker Image Created

```
root@ip-172-31-35-199:~/Brain-Tasks-App# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
brainstack    latest    41c0cb81c8d5   8 seconds ago  62.4MB
nginx         alpine    b76de378d572   10 days ago   62.1MB
root@ip-172-31-35-199:~/Brain-Tasks-App#
```

Run Docker Image

```
root@ip-172-31-35-199:~/Brain-Tasks-App# docker run -p 3000:3000 brainstack:latest
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
```

Verify the application running in Browser



ECR Repo Creation

```
root@ip-172-31-25-199:~# aws ecr create-repository \
--repository-name react-app \
--region ap-south-1
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:ap-south-1:166647428737:repository/react-app",
    "registryId": "166647428737",
    "repositoryName": "react-app",
    "repositoryUri": "166647428737.dkr.ecr.ap-south-1.amazonaws.com/react-app",
    "createdAt": "2026-02-15T06:22:08.794000+00:00",
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
```

Docket Image Tagged to ECR Repo name

```
root@ip-172-31-25-199:~# docker tag brainstack:latest 166647428737.dkr.ecr.ap-south-1.amazonaws.com/react-app:latest
root@ip-172-31-25-199:~# docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
brainstack	latest	41c0cb81c8d5	21 minutes ago	62.4MB
166647428737.dkr.ecr.ap-south-1.amazonaws.com/react-app	latest	41c0cb81c8d5	21 minutes ago	62.4MB
nginx	alpine	b76de378d572	10 days ago	62.1MB

```
root@ip-172-31-25-199:~#
```

Pushed Image to ECR

```

root@ip-172-31-35-199:~# docker push 166647428737.dkr.ecr.ap-south-1.amazonaws.com/react-app
Using default tag: latest
The push refers to repository [166647428737.dkr.ecr.ap-south-1.amazonaws.com/react-app]
6c266e2857ea: Pushed
58741a14e927: Pushed
dcb2431058a7: Pushed
da3ac26fd0f: Pushed
7b2903554e63: Pushed
6c0e59fd128a: Pushed
aa0fc249df10: Pushed
f35bcec50d8c: Pushed
660f9a93104f: Pushed
53998a5033c3: Pushed
989e799e6349: Pushed
latest: digest: sha256:ec2d73e038e5ff0247fda2573b6bde6aaffc31fcbd418235a5b19e96b815c865 size: 2613
root@ip-172-31-35-199:~#

```

Amazon Elastic Container Service

Private registry

Repositories

react-app

Features & Settings

Public registry

Repositories

Settings

react-app

Summary | Images | Lifecycle policy | Permissions | Repository tags

Images (1) Info

Filter active images

Image tags	Type	Created at	Image size (MB)
latest	Image	February 15, 2026, 12:05:26 (UTC+05.5)	26.08

EKS Cluster Created with 7 nodes

```

root@ip-172-31-35-199:~# eksctl create cluster \
--name react-eks-cluster \
--region ap-south-1 \
--nodegroup-name reactapp-nodes \
--node-type t3.micro \
--nodes 7
2026-02-15 06:40:05 [i] eksctl version 0.223.0
2026-02-15 06:40:05 [i] using region ap-south-1
2026-02-15 06:40:05 [i] setting availability zones to [ap-south-1a ap-south-1c ap-south-1b]
2026-02-15 06:40:05 [i] subnets for ap-south-1a - public:192.168.0.0/19 private:192.168.96.0/19
2026-02-15 06:40:05 [i] subnets for ap-south-1c - public:192.168.32.0/19 private:192.168.128.0/19
2026-02-15 06:40:05 [i] subnets for ap-south-1b - public:192.168.64.0/19 private:192.168.160.0/19
2026-02-15 06:40:05 [i] nodegroup "reactapp-nodes" will use "" [AmazonLinux2023/1.34]
2026-02-15 06:40:05 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed ne
t, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-02-15 06:40:05 [i] using Kubernetes version 1.34

```

Amazon Elastic Kubernetes Service

Dashboard

Clusters

Settings

Dashboard settings

Console settings

Clusters (1) Info

Filter clusters

Cluster name	Status	Kubernetes version	Support period
react-eks-cluster	Active	1.34 Upgrade now	Standard support until December 2, 2026

Deployment.yaml

```

root@ip-172-31-35-199:~# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 5
  selector:
    matchLabels:
      app: react-app
  template:
    metadata:
      labels:
        app: react-app
    spec:
      containers:
        - name: react-app
          image: 166647428737.dkr.ecr.ap-south-1.amazonaws.com/react-app
          ports:
            - containerPort: 3000

```

Service.yaml

```

root@ip-172-31-35-199:~# cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: react-app-service
spec:
  type: LoadBalancer
  selector:
    app: react-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 3000

```

Applied Deployment & Service.yaml

```

root@ip-172-31-35-199:~# kubectl apply -f deployment.yaml
deployment.apps/brain-tasks-deployment created
root@ip-172-31-35-199:~# kubectl apply -f service.yaml
service/react-app-service created
root@ip-172-31-35-199:~#

```

Verified the Nodes & Pods

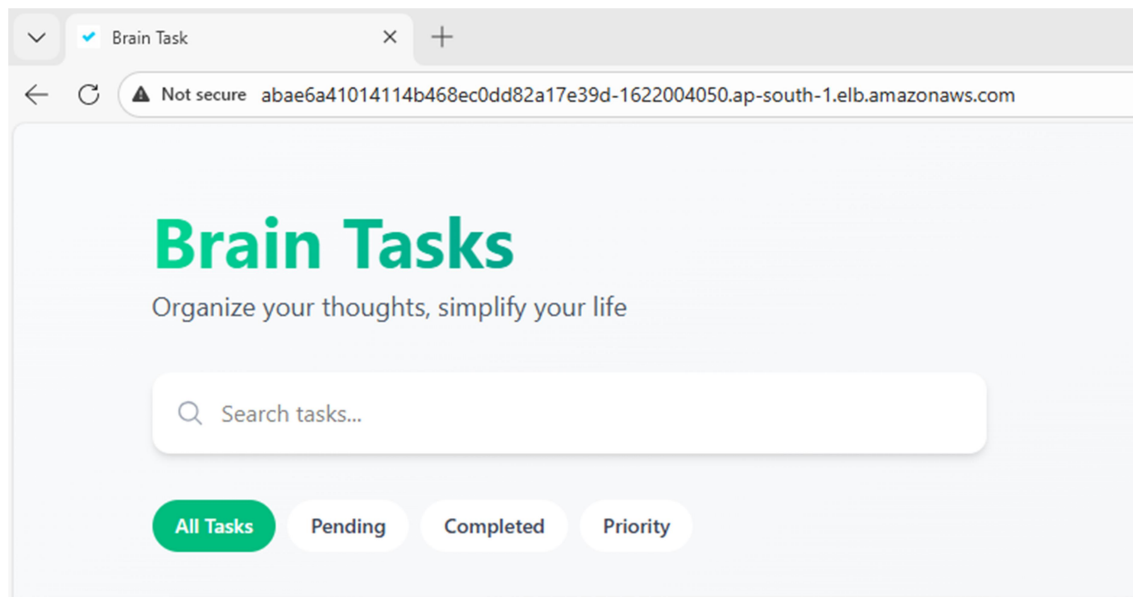
```

root@ip-172-31-35-199:~# kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-36-153.ap-south-1.compute.internal Ready    <none>   8m16s  v1.34.3-eks-70ce842
ip-192-168-39-8.ap-south-1.compute.internal Ready    <none>   8m17s  v1.34.3-eks-70ce842
ip-192-168-68-180.ap-south-1.compute.internal Ready    <none>   8m16s  v1.34.3-eks-70ce842
ip-192-168-72-45.ap-south-1.compute.internal Ready    <none>   8m16s  v1.34.3-eks-70ce842
ip-192-168-72-212.ap-south-1.compute.internal Ready    <none>   8m16s  v1.34.3-eks-70ce842
ip-192-168-9-247.ap-south-1.compute.internal Ready    <none>   8m16s  v1.34.3-eks-70ce842
ip-192-168-90-188.ap-south-1.compute.internal Ready    <none>   8m16s  v1.34.3-eks-70ce842
root@ip-172-31-35-199:~# kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
brain-tasks-deployment-7d5f577877-g6ctb 0/1      ImagePullBackOff 0             54s
brain-tasks-deployment-7d5f577877-37rps 0/1      ErrImagePull    0             54s
brain-tasks-deployment-7d5f577877-b6cph 0/1      ErrImagePull    0             54s
brain-tasks-deployment-7d5f577877-rdx4m 0/1      ImagePullBackOff 0             54s
brain-tasks-deployment-7d5f577877-s9nms 0/1      ErrImagePull    0             54s
root@ip-172-31-35-199:~# kubectl get svc
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes      ClusterIP   10.100.0.1     <none>          443/TCP    14m
react-app-service LoadBalancer 10.100.231.138 abae6a41014114b468ec0dd82a17e39d-1622004050.ap-south-1.elb.amazonaws.com 80:31294/TCP 51s

```

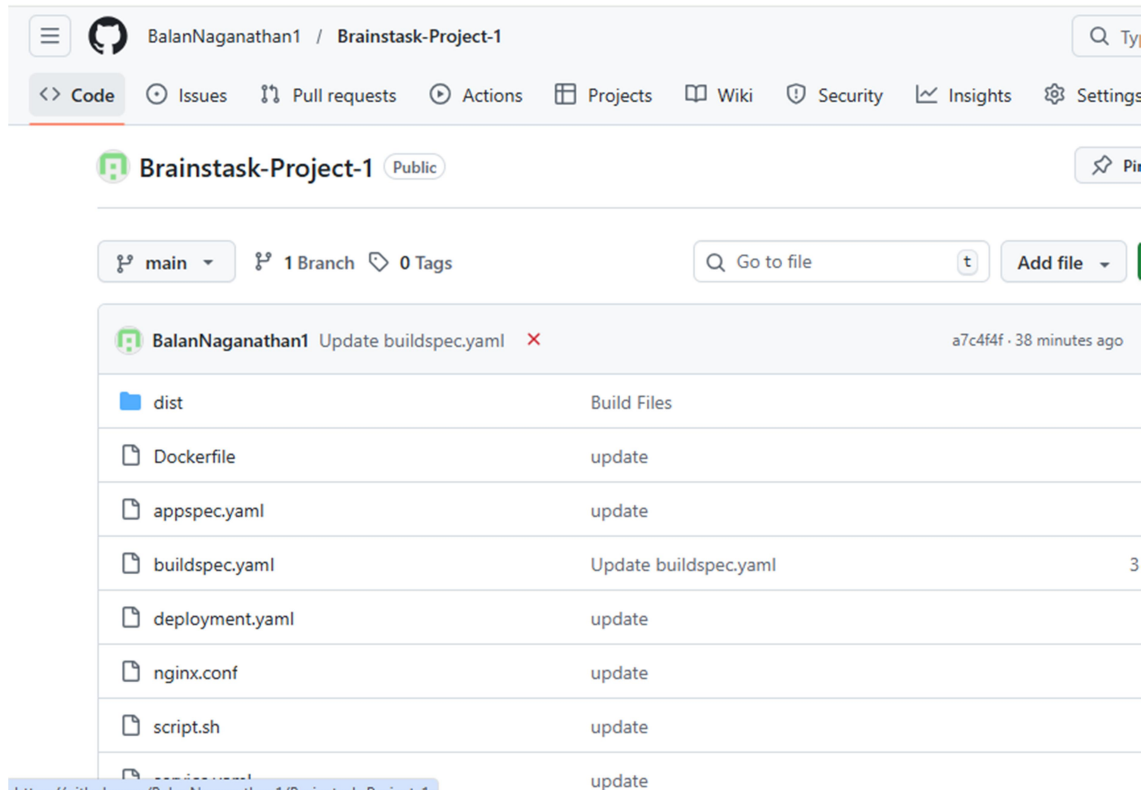
Verified app using Loadbalancer ARN

abae6a41014114b468ec0dd82a17e39d-1622004050.ap-south-1.elb.amazonaws.com



Pushed code to Github Repo

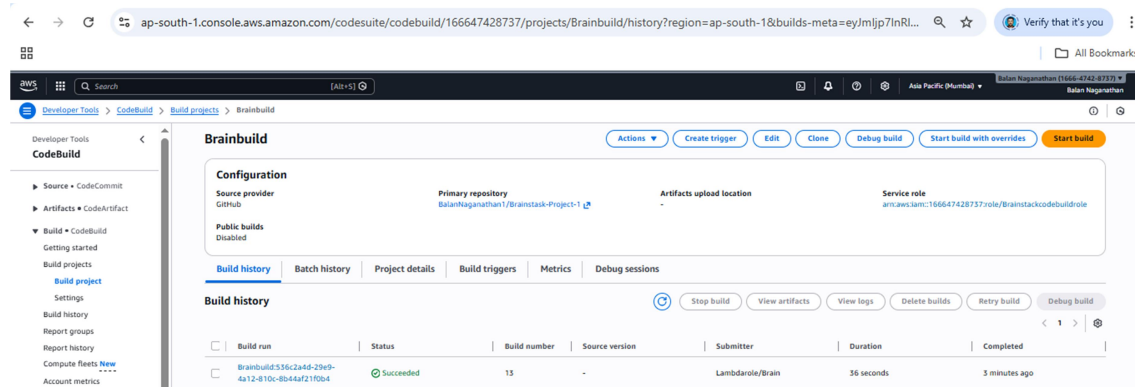
```
root@ip-172-31-35-199:~/Brain-Tasks-App# git push -u origin main
Username for 'https://github.com': BalanNaganathan1
Password for 'https://BalanNaganathan1@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.64 KiB | 1.64 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/BalanNaganathan1/BrainTask-Project-1.git
 f7dcd83..baa5099  main -> main
branch 'main' set up to track 'origin/main'.
root@ip-172-31-35-199:~/Brain-Tasks-App# |
```



Updated the Cluster Config

```
root@ip-172-31-35-199:~/Brain-Tasks-App# kubectl apply -f aws-auth.yaml
configmap/aws-auth unchanged
root@ip-172-31-35-199:~/Brain-Tasks-App# cat aws-auth.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: aws-auth
  namespace: kube-system
data:
  mapRoles: |
    - rolearn: arn:aws:iam::166647428737:role/Brainstackcodebuildrole
      username: codebuild
      groups:
        - system:bootstrappers
        - system:masters
root@ip-172-31-35-199:~/Brain-Tasks-App#
```

Build Executed in Codebuild using Lambda



Capture the Logs in Cloudwatch

